



HABIB
UNIVERSITY

Midterm Exam

Course: Electromagnetic Theory

Instructor: Dr. Shafayat Abrar

Date: Mar. 20, 2021

Marks: points

INSTRUCTIONS:

1. YOU ARE NOT ALLOWED TO DISCUSS WITH EACH OTHER.
2. DO NOT SHARE YOUR ROUGH/FAIR WORK WITH OR COPY/CHEAT FROM OTHERS; YOUR WORK MUST BE YOUR “INDEPENDENT” WORK.
3. NO PENCIL WORK, USE PROPER BLACK MARKER OR POINTER.
4. PROVIDE A NEAT WORK WHILE AVOIDING EXCESSIVE NUMBER OF PAGES.
5. ENCLOSE YOUR FINAL RESULTS IN RECTANGULAR BOXES.
6. SINGLE COLUMN ONLY IN PORTRAIT FORMAT. LEAVE SOME SPACES AS LEFT/RIGHT MARGINS.
7. IF REQUIRED, ADJUST CONTRAST LEVEL IN SCANS TO REMOVE DARK BACKGROUND.
8. IF REQUIRED, CROP YOUR IMAGES TO REMOVE UNNECESSARY SURROUNDINGS.

1. (20 points) Obtain the electric field intensity at a distance h above the center of a square loop (each side a) carrying a uniform line charge density ρ_L .

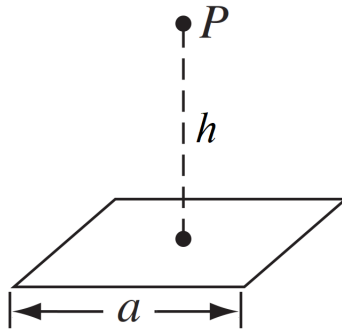


Figure 1: Figure for Q1.

2. (15 points) There are two parallel infinite planes; they carry equal but opposite uniform charge densities $\pm\rho_S$. Find the electric field in each of the three regions: (i) to the left of both, (ii) between them, (iii) to the right of both.

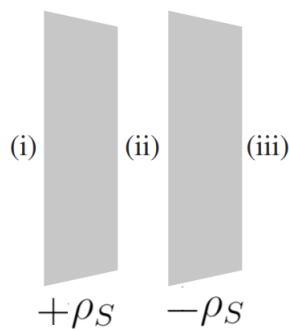


Figure 2: Figure for Q2.

3. (10 points) Use Gauss's law to find the electric field inside a uniformly charged solid sphere (the volumetric charge density, ρ_v and the radius, R).

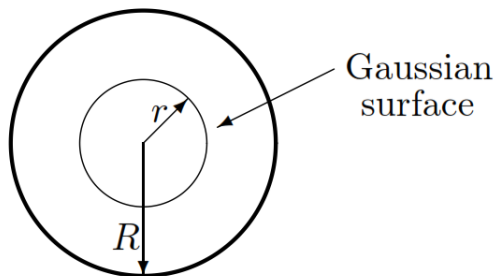


Figure 3: Figure for Q3.

4. (20 points) A thick spherical shell carries charge density

$$\rho_\nu = \frac{k}{r^3}, \quad a \leq r \leq b$$

Find the electric field in the three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$.

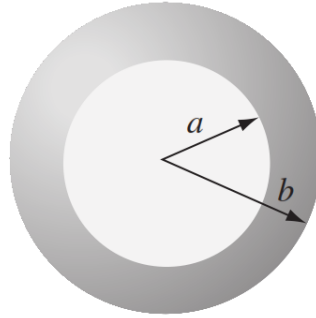


Figure 4: Figure for Q4.

5. (15 points) An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ_ν placed parallel to x-y plane such that the x-y plane passes through the middle of the slab. Find the electric field inside the slab.

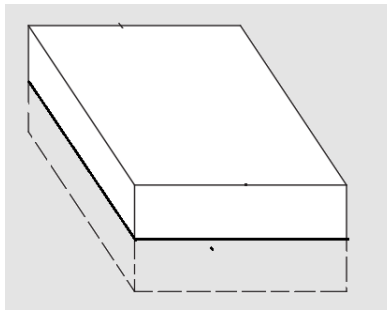


Figure 5: Figure for Q5.

6. (20 points) Consider a solid spherical shell of charge of radius R and the total charge q ; the electric field is given as

$$\mathbf{E} = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \mathbf{a}_r & \text{outside the sphere} \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \mathbf{a}_r & \text{inside the sphere} \end{cases}$$

Find the electric potential both inside and outside the sphere using infinity as the reference point ($V(\infty) = 0$).

7. (15 points) Three charges are situated at the corners of a square (side a), as shown below. **(a)** How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner? **(b)** How much work does it take to assemble the whole configuration of four charges?

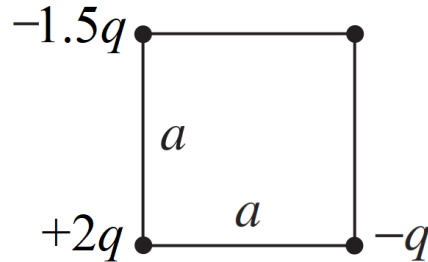


Figure 6: Figure for Q6.

8. (10 points) Two positive point charges, q_A and q_B (masses m_A and m_B) are at rest, held together by a massless string of length a . Now the string is cut, and the particles fly off in opposite directions. How fast is each one going?

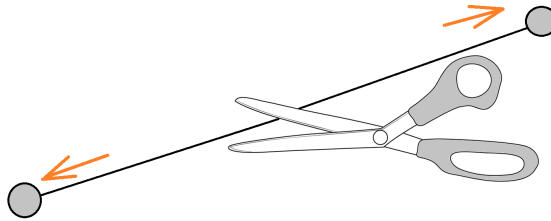


Figure 7: Figure for Q7.